

⇒ What is an array?

An array is a fundamental data structure used in programming to store a collection of items the same data type in a single variable.

Instead of creating separate variables for every single item, an array lets you group them together under one name in a fixed, contiguous sequence.

Key concepts: Zero-Based Indexing: in almost all programming languages, arrays start counting positions from 0, not 1.

• Contiguous Memory: All elements are stored side-by-side in computer memory, making it extremely fast to access any item directly if you know its index.

When should you use an array?

- Fixed-size lists.
- Fast access.
- Sequential processing

ex. Python:

```
scores = [85, 92, 78, 90, 88]
```